UNIVERSITY EXAMINATIONS: 2022/2023

EXAMINATION FOR BACHELOR OF EDUCATION

SMA 2002/ECON 110: CALCULUS I

DATE: APRIL, 2023 TIME: 2 HOURS

INSTRUCTIONS: Answer Question One and Any Other Two Questions

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QUESTION ONE (20 MARKS)

(a) Let $g: \mathbb{R} \rightarrow \mathbb{R}$ where $\mathbb{R}$ is a set of real numbers such that $g(x) = x^2 + x - 6$, find the $g^{-1}(x)$. **(4 Marks)**

(b) Let $y = uv$ be the product of functions u and v. Find $y'(2)$ if $u(2) = 2, u'(2) = -5$, $v(2) = 1$ and $v'(2) = 3$. **(2 Marks)**

(c) Evaluate $\lim_{x \to 1} \frac{x^2+x-2}{x^2-x}$ **(2 Marks)**

(d) Given $f(x) = (x + 1)^3$, find the slope of the normal line at the point (-2,-1). **(4 Marks)**

(e) Find the derivative of the function $y = \frac{e^{-3xy}}{x^2+1}$ **(4 Marks)**

(f) If $x = t^3$, $y = t^2 - t$, find $\frac{d^2y}{dx^2}$. **(4 Marks)**

QUESTION TWO (15 MARKS)

(a) If $\frac{\mathrm{d}\,(\sin x)}{\mathrm{dx}} = \cos x$, $\frac{\mathrm{d}(\cos x)}{\mathrm{dx}} = \sin x$ derive the expressions for

(i) $\frac{\mathrm{d}\,(\tan\ x)}{\mathrm{dx}}$ **(3 Marks)**

(ii) $\frac{\mathrm{d}\,(\sin^{-1} x)}{\mathrm{dx}}$ **(4 Marks)**

(b) Given that $y = \frac{\cos x}{x}$, prove that $\frac{d^2y}{dx^2} + \frac{2}{x}\frac{dy}{dx} + y = 0$. **(5 Marks)**

(c) Differentiate with respect to x, the function $y = sec^2 2x$. **(3 Marks)**

QUESTION THREE (15 MARKS)

(a) Show that $f(x) = \frac{x^2+x-6}{x^2-4}$, has a removable discontinuity at x=2. **(3 Marks)**

(b) If $y = \sqrt{\frac{1+\cos x}{1-\cos x}}$, show that $\frac{dy}{dx} = \frac{-1}{1 - \cos x}$

Hint: $\sin x = \sqrt{1 - cos^2 x}$ **(4 Marks)**

(c) Find the derivative of $f(x) = \sqrt{x}$ using the method of differentiation from first principles.

(4 Marks)

(d) Let $f: \mathbb{R} \to \mathbb{R}$ and $g: \mathbb{R} \to \mathbb{R}$, where $\mathbb{R}$ is a set of real numbers. If $f(x) = x^2 - 2$ and $g(x) = x + 4$, find $(fog)^{-1}x$. **(4 Marks)**

QUESTION FOUR (15 MARKS)

(a) The distance x metres moved by a car in time t seconds is given by $x = 3t^3 - 2t^2 + 4t - 1$. Determine the velocity and acceleration when $t = 1.5$ **(4 Marks)**

(b) Using $y = \sqrt[3]{x}$, find the approximate value of $\sqrt[3]{124}$. **(4 Marks)**

(c) The parametric equations of a curve are $x = e^t, y = \sin t$. Find $\frac{d^2y}{dx^2}$ as a function of t. Hence show that $x^2 \frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = 0$. **(7 Marks)**

QUESTION FIVE (15 MARKS)

(a) If $y^3 - 3yx^2 + 2x^3 = 0$, find $\frac{dy}{dx}$. **(4 Marks)**

(b) Show that the rectangle with area $100\ m^2$ and minimum perimeter is a square. **(4 marks)**

(c) Find the turning points and the points of inflection on the curve $y = x^5 - 5x^4 + 5x^3 - 1$. **(7 Marks)**